

Face Unlock App — Documentation

Methodology

This project implements a minimal face recognition-based authentication system using the DeepFace library and Streamlit. The goal was to create a simple application that allows a user to:

1. Register their face with a name.
2. Login by comparing their live face with registered data.

Main Steps:

- Capture image from webcam via OpenCV.
- Save the face image locally with a user-defined name.
- Compare the login face with all saved images using DeepFace's `verify()` function with the FaceNet model.

Dataset Preparation

The system dynamically builds a small face dataset by:

- Capturing images from the webcam during registration.
- Saving them under `registered_faces/` with the filename as the user's name.

- During login, captured face is compared against all saved images.

Choice of Architecture

Face Recognition Model: FaceNet

Tools Used:

- DeepFace
- OpenCV
- Streamlit
- Python(3.11)

Experimental Results

Test Cases:

Register face and match login	Success
Try unknown/unregistered face	Access Denied
Face partially blocked	Sometimes Fails
Low lighting condition	Low accuracy

Potential Real-World Applications

This kind of face recognition system can be used in:

- Personal device security (Face Unlock).
- Attendance systems (Workplace).
- Entry gate automation (Offices).

- IoT home security (unlock door via camera).